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ORIGINAL ARTICLE

Genome-wide identification, in-silico characterisation and expression analysis of multiprotein bridging factor 1 gene family members in rice

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Abstract

The multiprotein bridging factor 1 (MBF1) proteins are evolutionarily conserved transcription co-factors. However, little is known about rice *MBF1* gene family and its role. A genome-wide search led to the identification of two *MBF1* genes in the rice genome. Their proteins contained characteristic MBF1 and helix-turn-helix domains. Phylogenetic analysis showed that they belong to two different groups. Exploration of publicly available rice transcriptome data revealed that *OsMBF1b* exhibits constitutively high transcript abundance in all tissues and developmental stages of rice with a little alteration in its expression. Contrarily, *OsMBF1c* exhibited a prominent alteration in its expression in response to environmental perturbations. Both *OsMBF1s* showed the highest expression in endosperm. Analysis of publicly-available rice transcriptome data also showed that both *OsMBF1s* have a role in response to different stresses, especially in heat. Transcript analysis using qRT-PCR confirmed heat inducibility of *OsMBF1c* in contrasting genotypes i.e. IR64 (heat sensitive) and Nagina 22 (heat tolerant). qRT-PCR also confirmed the drought inducibility of both genes in the IR64 genotype which is sensitive to drought stress also as revealed by analysis of different parameters. In-silico interaction study also indicated their role in heat response as a number of proteins required to cope with high temperatures were predicated to be their interacting partners. Several heat-responsive genes were found to co-express with *OsMBF1s*. In-silico promoter analysis revealed the occurrence of stress-responsive elements in their putative promoters. Interestingly, both *OsMBF1s* showed diurnal rhythmic expressions having peaks during the daytime when the temperature rises. Altogether, this study indicates an active role of *OsMBF1s* in thermotolerance in rice. This is the first report regarding the characterization of rice *MBF1* members.

Keywords Multiprotein bridging factor 1 · Rice · Heat · Interaction · qRT-PCR

Introduction

The multiprotein bridging factor 1 (MBF1) proteins are found in all domains of life on earth from archaea to eukaryotes (de Koning et al., 2009). In animals, they are also named as endothelial differentiation-related factor 1. MBF1 was first identified in silkworm (Li et al., 1994). It was found to be a part of the complex consisting of TATA box protein-associated factors (Li et al., 1994). It functions as a cofactor and forms a bridge between a transcription factor and the TATA box-binding protein (TBP). In yeast, it mediates interaction of GCN4 transcription factor with TBP (Takemaru et al., 1998). A few *MBF1* genes have been identified and characterized in plants (Tsuda et al., 2004; Yan et al., 2014; Zhang et al., 2020; Tian et al., 2022; Yu et al., 2021). In tomato, ethylene treatment induces *SlMBF1* expression in fruit (Hommel et al., 2008). *CaMBF1*, isolated from pepper (*Capsicum annuum*), negatively regulates cold response (Guo et al., 2014). In Chrysanthemum, overexpression of *DgMBF1* isolated from *Dendranthema grandiflorum* enhances salinity tolerance (Zhao et al., 2019). *Arabidopsis* has three *MBF1* genes i.e. *AtMBF1a* (At2g42680), *AtMBF1b* (At3g58680), and *AtMBF1c* (At3g24500) which respond to various stresses (Tsuda & Yamazaki, 2004; Arce et al., 2010). Overexpression of *AtMBF1a* imparts tolerance to salt stress and pathogens (Kim et al., 2007) and high

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temperature (Suzuki et al., 2005, 2008). Even heterologous expression of *MBF1s* from different sources confer tolerance to heat (Qin et al., 2015), drought (Yan et al., 2014), and salinity (Alavilli et al., 2017; Zhang et al., 2020) in *Arabidopsis*. Ectopic expression of a wheat *MBF1* is reported to improve heat tolerance in rice (Qin et al., 2015). There are two transcriptomic studies in which rice *MBF1* genes have been reported regarding their transcript abundance (Wang et al., 2010; Ray et al., 2011). In one of them, a rice *MBF1* (LOC_Os06g39240) was found to be induced by drought (Ray et al., 2011) while in the second study, another rice *MBF1* (LOC_Os08g27850) gene was found to possess high transcript abundance in all tissues (Wang et al., 2010). Till date, there is no other report regarding the characterization of rice *MBF1* genes. Rice is an important crop with respect to world food security. Different biotic and abiotic stressors threaten its production and cultivation. Therefore, there is a need to adopt suitable approaches to enhance its stress tolerance (Jangir et al., 2021; Shekhawat et al., 2021; Bishnoi et al., 2022; Priya et al., 2023). A better understanding of the underlying molecular mechanism is also essential. To test whether *MBF1* gene family has a role in rice immunity and stress tolerance, a bioinformatics investigation was carried out and publicly available rice transcriptome data was explored. Since a proper investigation regarding the number of *MBF1* genes in rice is not available; a systematic search was done and the whole rice genome was mined to identify the members of this family in rice. A total of two members were identified. Their structural and evolutionary relationships were analysed. Their responses to different environmental stresses were investigated using Genevestigator. Their tissue specificity, diurnal, and development-specific expression patterns were also determined. In addition, in-silico promoter analysis, co-expression analysis, and protein-protein interaction prediction were also carried out to understand their functional diversity. Their expression levels were analysed in IR64 rice seedlings subjected to different abiotic stresses. Various stress related parameters were also estimated to assess the level of stress resistance or susceptibility of this rice genotype. Their transcript levels were also checked in IR64 and Nagina 22 genotypes in response to heat stress. Overall, the present study improved the understanding of the potential role of *OsMBF1* genes and laid the foundation for further studies on their functional validation.

Methods

Identification of MBF1 family members in rice

Three independent approaches were used to identify *OsMBF1* members in rice; HMM profile search, BLASTP, and keyword/name search. The hidden Markov model (HMM) profiles of the MBF1 and HLH3 domains (Pfam IDs PF08523 and PF01381, respectively) were retrieved from the Pfam V35.0 database (http://pfam.xfam.org/). Then, these two HMM profiles were used as queries and BLAST searched against the protein sequences of rice in MSU Rice Genome Annotation Project (RGAP) Release 7 (http://rice.plantbiol-ogy.msu.edu/) database using default parameters. The protein sequences of all three AtMBF1s (as reported by Tsuda et al., 2004) were retrieved from TAIR database (https://www.arabidopsis.org) and searched individually by BLASTP in RGAP database with default parameters. A keyword search was carried out using the aforementioned Pfam IDs in the same database. 'Endothelial differentiation-related factor 1' name was also searched. After conducting all searches, obtained gene locus IDs were pooled together and redundant IDs were eliminated. Presence of MBF1 and HTH_3 domains were reconfirmed in the resultant unique proteins using the Pfam database. Names of identified members were coined based on their similarity to *Arabidopsis* MBF1 proteins. Additionally, the Rice Annotation Project (RAP) (https://rapdb.dna.affrc.go.jp/) database was also explored using the above-mentioned three approaches.

Analysis of gene structure and chromosomal location

Chromosomal coordinates, gene orientation and sequence (CDS and genomic) information of identified *OsMBF1* genes were obtained from the RGAP database. Molecular weight and theoretical isoelectric point (pI) of OsMBF1 proteins were calculated using the ExPASy server's Compute pI/ Mw tool (http://web.expasy.org/compute_pi/; Gasteiger et al., 2005). Map tool available at Oryzabase (http://viewer.shigen.info/oryzavw/maptool/MapTool.do) was used for creating chromosomal location map (Kurata & Yamazaki, 2006). By comparing genomic sequences with their corresponding coding sequences, the gene structures of *OsMBF1s* were drawn using Gene Structure Display Server v2.0 (http://gsds.gao-lab.org/; Hu et al., 2015).

Phylogenetic analysis

MBF1 protein sequences of *Gossypium hirsutum*, *Physcomitrium patens*, *Sorghum bicolor*, *Chlamydomonas reinhardtii*, *Panicum virgatum*, *Setaria italica*, *Solanum tuberosum*, and *Triticum aestivum* were obtained from Phytozome v13 (https://phytozome-next.jgi.doe.gov/) (Goodstein et al., 2012). For this, the keyword 'MBF1' was searched and the above-mentioned genomes were selected in the Phytozome v13 database. Sequences of MBF1 members of *Arabidopsis*, rice, and *S. cerevisiae* were downloaded from TAIR, RGAP, and *Saccharomyces* Genome Database (SGD, https://www.yeastgenome.org). Multiple sequence alignment of amino acid sequences of total 38 MBF1 proteins from these eleven species was carried out using MUSCLE in MEGA11 (Molecular Evolutionary Genetics Analysis Version 11; Edgar, 2004; Tamura et al., 2021) and based on the alignment file, the Neighbor-Joining phylogeny tree was built with 1000 bootstrap iterations.

Analysis of protein structure

Different databases such as Pfam (http://pfam.xfam.org/), SMART (http://smart.embl-heidelberg.de/), and Prosite (https://prosite.expasy.org/scanprosite/) were used to identify the domain boundaries in identified OsMBF1 proteins. Their secondary structures were predicted using PSIPRED protein structure prediction server (http://bioinf.cs.ucl.ac.uk/psipred/) (McGuffin et al., 2000). AlphaFold Protein Structure Database (AlphaFold DB, https://alphafold.ebi.ac.uk) was also used for the prediction of protein structure with a 3D structural model (Varadi et al., 2022).

Subcellular localization prediction

WoLFPSORT (https://wolfpsort.hgc.jp/; Horton et al., 2007) and LocTree3 (https://www.rostlab.org/services/loctree3/; Goldberg et al., 2014) were used for the prediction of subcellular localization of OsMBF1 proteins.

In-silico expression analysis

Genevestigator (https://genevestigator.com; Hruz et al., 2008) was used to analyse the transcript profiles of *OsMBF1* genes as regulated by tissue specificity, developmental stages, and different stresses. Co-expression analysis was also done using Genevestigator. Rice Affymetrix microarray data (Affymetrix Rice Genome array) as well as RNA seq (mRNA-seq Gene Level *Oryza sativa*) data were accessed. In the case of the former one, a single specific probe was found for each *OsMBF1* gene (Os.110.46. 1.S1_At and Os.7897.46. 1.S1_At for LOC_Os06g39240 (*OsMBF1c*) and LOC_Os08g27850 (*OsMBF1b*), respectively). In case of stresses (perturbation), differential expression was accessed with a filter of fold change with a cut-off value $\log_2$ FC > 1.5 and *P*-value < 0.05. To analyse temporal expression patterns of *OsMBF1*s, Diurnal (http://diurnal.mocklerlab.org/) was used (Mockler et al., 2007). It has integrated data from time-dependent microarray experiments carried out on model plants under different conditions. Best-fitting models correlated with the temporal patterns of each *OsMBF1* were also accessed using it. The detailed experimental conditions regarding rice circadian transcriptomes (the data of which is incorporated in Diurnal) is mentioned in the publication by Filichkin et al. (2011).

Interaction analysis

Protein-protein interaction was predicted by exploring STRING database v11.5 (https://string-db.org/; Szklarczyk et al., 2017). Full STRING network, as well as physical subnetwork, were accessed for each OsMBF1 protein with the setting of required interaction with high confidence (score > 0.7) and size cut off no more than 10 interactors. Then, protein sequences of identified interactors were downloaded and their corresponding locus IDs were identified by BLASTP analysis in RGAP database.

In-silico promoter analysis

A search for regulatory elements in the putative promoters of each *OsMBF1* gene was performed using the PlantCARE (Plant Cis-Acting Regulatory Element) database (https://bioinformatics.psb.ugent.be/webtools/plantcare/html/; Lescot et al., 2002) and ppdb (Plant Promoter Database v3.0) database (Hieno et al., 2014). While exploring ppdb, their RAP (The Rice Annotation Project; https://rapdb.dna.affrc.go.jp/) IDs were used to identify their corresponding cDNA accessions. RAP IDs of *OsMBF1c* and *OsMBF1b* are Os06g0592500 and Os08g0366100, respectively. 550 bp genomic sequence upstream of TSS as well as 5'UTR sequence of each *OsMBF1* gene was retrieved from RGAP database and subsequently scanned by PlantCARE.

Transposon insertion analysis

RTRIP (Rice Transposons Insertion Polymorphism) database (http://ibi.zju.edu.cn/Rtrip/index.html; Liu et al., 2020) was accessed to detect transposon insertion in *OsMBF1* genes.

Plant growth and stress treatment

Rice seeds (IR64) were surface sterilized and germinated on a threefold layer of Whatman No. 1 filter paper in 90 mm Petri dishes at 28 ± 2 °C. Germinated seeds were transferred to a hydroponics system having half-strength Yoshida medium for further growth and development in a plant growth chamber (Conviron CMP 6050) at phytotron facility at NIPGR, New Delhi, India. It maintained 28 ± 2 °C Day/25 ± 2 °C night temperature with 80 ± 5% relative humidity, light 400µmolm$-^2$s$-^1$ and 16 h/8 h LD photoperiod (lights on from 6 am to 10 pm). 10 days after the transfer, seedlings were subjected to salt, heat, and drought stress. Salt stress was imposed using 200mM NaCl in half-strength Yoshida medium. 25% PEG-6000 was used to induce drought stress. One set of seedlings was shifted to another growth chamber set at 42°C with all other parameters similar to the control chamber. After 6 h of stress treatment, leaf tissue (youngest leaf) was harvested and flash frozen in liquid nitrogen and stored at -80°C until total RNA was isolated for gene expression analysis. Leaf tissue of non-stressed seedlings (control) was also harvested at the same time. Stress-associated growth and physiological parameters were analyzed after 30 h of stress treatment when stress phenotype was visible.

As bioinformatics analyses strongly suggested a prominent role of *OsMBF1s* in heat response in rice, we extended analysis of their expression patterns in contrasting genotypes (Nagina-22, heat tolerant and IR64, heat susceptible) for thermotolerance. For this, a separate experiment was conducted. Seeds were sown in soilrite in the above-mentioned growth chamber with the same growth conditions. A total of three sets (for control, heat treatment, and heat recovery) were maintained for each genotype. After twenty-one days of germination, two sets of rice seedlings of each genotype were subjected to heat stress. Heat stress was applied at 44°C for 12 h (8:00 a.m. to 8:00 p.m.) using the growth chamber with all other parameters mentioned above. At the end of the heat stress treatment (12 h), leaf tissue was harvested, flash-frozen in liquid nitrogen and stored at -80°C. Leaf tissue from untreated seedlings i.e. control seedlings was also harvested and stored as mentioned earlier. After heat treatment, one set of heat-stressed rice seedlings was transferred to control conditions for 24 h recovery and subsequently tissue was harvested on the next day. Leaves from three seedlings were pooled to constitute a biological replicate. Total five biological replicates were used to check the expression of the *OsMBF1* family genes.

Primer designing

The cDNA sequences of rice MBF1 gene family members were retrieved from RAP (The Rice Annotation Project; https://rapdb.dna.affrc.go.jp/) and used to design primers for qRT-PCR (Table 1). The last exon and 3' UTR region were used for primer designing. Primers were designed using Primer 3.0 (version 4.1.0) with the following parameters: 20 ± 3 nucleotides length; 50–120 bp amplicon size; 57 to 63°C melting temperature (Tm); 50 ± 5% GC content with no secondary structures.

Table 1 Details of primers of *OsMBF1* genes

Gene name/ Locus ID	Sequence (5′–3′)	Amplicon size
OsMBF1c LOC_Os06g39240	F- CGCCAAGTGAATCCTTCGTG R- AACTAGGAGGGATCGGAGGT	51 bp
OsMBF1b LOC_Os08g27850	F- AACCAGCAGATCATCGGGAA R- AACACTATGCTTCAGGGCCT	95 bp

Isolation of total RNA and cDNA synthesis

TRI Reagent® (Sigma-Aldrich) was used to isolate total RNA. NanoDrop-1000 spectrophotometer (Thermo Scientific) was used to determine RNA concentration. Integrity of each RNA sample was also verified by electrophoresis. To remove DNA contamination, total RNA was treated with DNase (TURBO DNA-free™ Kit, Invitrogen). The first-strand cDNA synthesis was carried out using 2 µg of DNA-free total RNA using iScript™ cDNA Synthesis Kit (Bio-Rad) in a final reaction volume of 40 µl following manufacturer's instructions and was stored at −20 °C until further use.

qRT-PCR and data analysis

Quantitative Real-time PCR was performed using QuantStudio™ 6 Flex Real-Time PCR System (Applied Biosystems). Each 10 µl qRT-PCR reaction mixture had 2 µl of 5-fold diluted cDNA, 10µM of each primer and 5 µl of 2× Power SYBR Green PCR Master Mix (Applied Biosystems, USA). *OseEF1α* was used as a reference gene. The qRT-PCR cycling conditions were as follows: an initial hold at 50°C for 2 min, an initial denaturation step at 95°C for 2 min, and 40 cycles of 10s at 95°C (denaturation), and 30s at 60°C (annealing cum extension) followed by melt curve analysis using default parameters. Five biological replicates of each sample and two technical replicates of each biological sample were used. Relative expression levels of both genes were calculated using delta-delta CT-method (ΔΔCT) (Livak & Schmittgen, 2001). One-way ANOVA was applied to check significant differences among means of control and different stresses. Further, Tukey's post-hoc test ($P = 0.05$) was conducted to compare these mean values using GraphPad Prism 9.5.1.733 software version 2.0. Graphical representation of qRT-PCR results was also executed using this software.

Measurement of different parameters of IR64 rice seedlings

Following parameters were analyzed after 30 h of treatment of the abovementioned stresses to IR64 rice seedlings:

Measurement of shoot length, root length, fresh weight and dry weight

Shoot and root length were measured using a meter scale. After wiping the roots of the seedlings with tissue paper, their fresh weight (FW) was measured. After drying the samples in an oven at 70°C for 72 h, their dry weight (DW) was measured. Data from 15 biological replicates was recorded for these parameters.

Estimation of relative water content

Relative water content (RWC) of seedlings was measured according to Lutts (1996). Briefly, the FW of each seedling was recorded immediately after harvesting as mentioned earlier. Its turgid weight (TW) was measured following incubation in water for 6 h. Following the determination of TW, seedlings were completely dried in an oven to measure DW. RWC from 15 biological replicates was calculated using the following formula: RWC = [(FW-DW)/(TW-DW)] X100 (%).

Estimation of electrolyte leakage

Electrical leakage (EL) was measured following Taratima et al. (2022) with minor modifications. Briefly, 200 mg of leaf tissue was collected from the fully expanded youngest leaf of seedlings. The leaf segments were thoroughly washed with Milli-Q water and placed in 20mL Milli-Q water. They were incubated at 28°C. After 12 h of incubation, initial electrical conductivity (EC1) was recorded using a conductivity meter (LMCM20, Labman Scientific Instruments). After autoclaving for 20 min at 121°C and 15 psi, total electrical conductivity (EC2) was measured. %EL was calculated as: (EC1/EC2) X 100%. Five biological replicates were taken for electrolyte leakage.

Estimation of photosynthetic pigments

Estimation of photosynthetic pigments was done following Arnon (1949). 200 mg of fresh rice leaf sample was incubated in 10mL of 80% acetone for 48 h to allow complete pigment extraction, and the solution was centrifuged at 12,000 g for 10 min. The absorbance of the supernatant was measured at 470, 645, 646, and 663 nm using 80% acetone as blank. Calculations for contents of chlorophyll a, b, total chlorophyll (Arnon, 1949), and carotenoid (Lichtenthaler, 1983) were done. Ten biological replicates were taken.

Statistical analysis of data of different parameters

Statistical tool GraphPad Prism 9.5.1.733 software version 2.0 was used for this purpose. Outliers were identified using this tool and removed from the obtained data. The normal distribution of the data was checked. One-way analysis of variance (ANOVA) was carried out. It was followed by multiple comparisons of mean values using Tukey's test where standard deviations (SDs) of means were not significantly different. Alternatively, where SDs were found to be significantly different, Brown-Forsythe and Welch ANOVA with Games Howell multiple comparison test was performed to identify significant differences between control and treatment groups.

Results

Identification of rice MBF1 members

Oryza sativa Nipponbare reference genome (MSU Rice Genome Annotation Project Release 7) was mined using three strategies (1) BLASTP, (2) HMM profile search, and (3) name search. *Arabidopsis* contains three *AtMBF1* genes with gene model IDs At2g42680, At3g58680 and At3g24500. Their protein sequences were retrieved from TAIR (https://www.arabidopsis.org/) and used as queries for BLASTP in RGAP database with default parameters. Query coverage varied from 93 to 100% and identity was found to be 49 to 81%. E values were less than $1e^{-20}$. It resulted in the identification of only two members of the *MBF1* family in the rice genome with locus IDs LOC_Os06g39240 and LOC_Os08g27850. Since already characterized MBF1 proteins contain two characteristic domains known as multiprotein bridging factor 1 (MBF1) and helix-turn-helix (HTH_3) having Pfam accession numbers PF08523 and PF01381, respectively; therefore, HMM profiles of these two accession numbers were obtained from Pfam database and searched through BLASTP in the RGAP database to identify rice proteins containing these two domains. 'Endothelial differentiation-related factor 1' keyword and names of the aforementioned domains were also searched. These searches also led to the identification of the same two genes as mentioned above. The Rice Annotation Project (RAP) (https://rapdb.dna.affrc.go.jp/) was also searched using the aforementioned three approaches and it also resulted in the identification of the same two genes. Rice MBF1 with locus ID LOC_Os06g39240 showed homology with AtMBF1c (At3g24500, E value $1e^{-44}$, Identity 60%, Positive 73%, Gap 3%). The second one with locus ID LOC_Os08g27850 exhibited homology with other two *Arabidopsis* MBF1s i.e., AtMBF1a (At2g42680, E value $3e^{-83}$, Identity 82%, Positive

92%, Gap 0%) as well as AtMBF1b (At3g58680, E value 1e^{-82}, Identity 80%, Positive 94%, Gap 0%). Therefore, rice *MBF1* genes were named *OsMBF1c* (LOC_Os06g39240) and *OsMBF1b* (LOC_Os08g27850), respectively according to the names of their *Arabidopsis* homologs.

Gene structure, protein structure and phylogenetic analysis

OsMBF1c and *OsMBF1b* were found to be localized on chromosomes 6 and 8, respectively. Their physical location map was drawn (Fig. 1). Their gene IDs, genomic positions, alternative splice forms, molecular weight of proteins, etc. were summarized (Table S1). *OsMBF1c* (LOC_Os06g39240) gene was found to contain no intron while *OsMBF1b* (LOC_Os08g27850) contained three introns (Fig. 1). These two genes encode proteins containing 142 and 155 amino acids, respectively. The predicted molecular weight and pI of OsMBF1b were observed to be 16.15 kDa and 10.67, respectively while OsMBF1b had a 15.62 kDa molecular weight with pI 9.95.

To explore the phylogenetic relationships of OsMBF1s, amino acid sequences of 38 MBF1s from eleven species namely, *Gossypium hirsutum*, *Oryza sativa*, *Arabidopsis thaliana*, *Physcomitrium patens*, *Sorghum bicolor*, *Chlamydomonas reinhardtii*, *Panicum virgatum*, *Setaria italica*, *Solanum tuberosum*, *Triticum aestivum* along with that of *Saccharomyces cerevisiae* were included for data analysis. Protein sequences of MBF1s of *Arabidopsis*, rice, and *S. cerevisiae* were downloaded from TAIR, RGAP, and *Saccharomyces* Genome Database (SGD, https://www.yeastgenome.org), respectively. Sequences of MBF1 of other plant species were retrieved from Phytozome v13 (https://phytozome.jgi.doe.gov/pz/portal.html). A phylogenetic tree was built from the alignment of amino acid sequences of full-length protein sequences using the neighbor-joining method in MEGA11 program (Fig. 2). As shown in Fig. 2, MBF1s got clustered into two major groups having subgroups. Two OsMBF1s belonged to two separate groups. Group I had 15 proteins including OsMBF1c, AtMBF1c (At3g24500), and YOR298C-A from *S. cerevisiae*. The second group contained 23 proteins including OsMBF1b, AtMBF1b (At3g58680), and AtMBF1a (At2g42680). This result was consistent with the similarity found between *Arabidopsis* and rice MBF1 homologs during BLAST search as mentioned before. The result also indicated an increase in the number of group II members in *Gossypium hirsutum*, *Solanum tuberosum,* and *Triticum aestivum.*

A representative multiple sequence alignment of plant MBF1 proteins along its phylogenetic tree revealed the presence of similar and identical residues, especially in the regions of two conserved domains (Fig. 3a and 3b). Both OsMBF1 proteins contained MBF1 and HTH_3 domains at N and C terminus, respectively (Fig. 3c). Secondary structure analysis using PSIPRED (protein structure prediction server) (http://bioinf.cs.ucl.ac.uk/psipred; McGuffin et al., 2000) showed the presence of helixes and coils in the C-terminus region while the N-terminal region also contains strands (Fig. 3d). OsMBF1c protein with 155 amino acids, was calculated to have 38.7% α-helices, 4.52% extended strands and 61.29% random coils. OsMBF1b contained 3.52% extended strands, 37.32% α-helices and 59.15% random coils.

Structural models of OsMBF1 proteins as predicted by AlphaFold also supported the observations of PSIPRED (Supplementary Material Fig. S1). Structure of both OsMBF1s at the N-terminal was extended while C-terminal had a well-folded orientation (Fig. S1). At the N terminal half, both of them contain 4 helixes. The prediction of sub-cellular localization by LocTREE3 tool showed that both OsMBF1 proteins are localized in the nucleus. However, WOLFPSORT tool predicted that OsMBF1c is cytoplasmic while OsMBF1b is a nuclear protein.

In-silico expression analysis based on publicly available transcriptome data

Tissue-specific and development phase-specific expression analysis

Genevestigator tool was used to explore the expression profile of each *OsMBF1* gene. *O. sativa* transcriptome

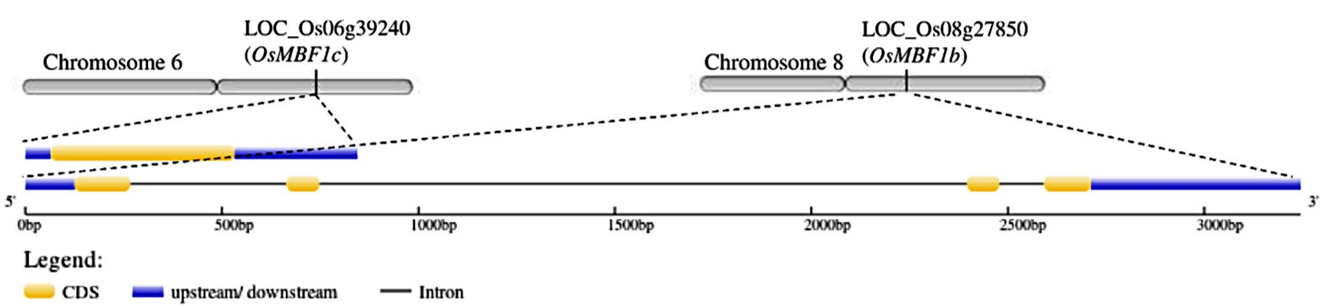

Fig. 1 Chromosomal location and structure of *OsMBF1* genes

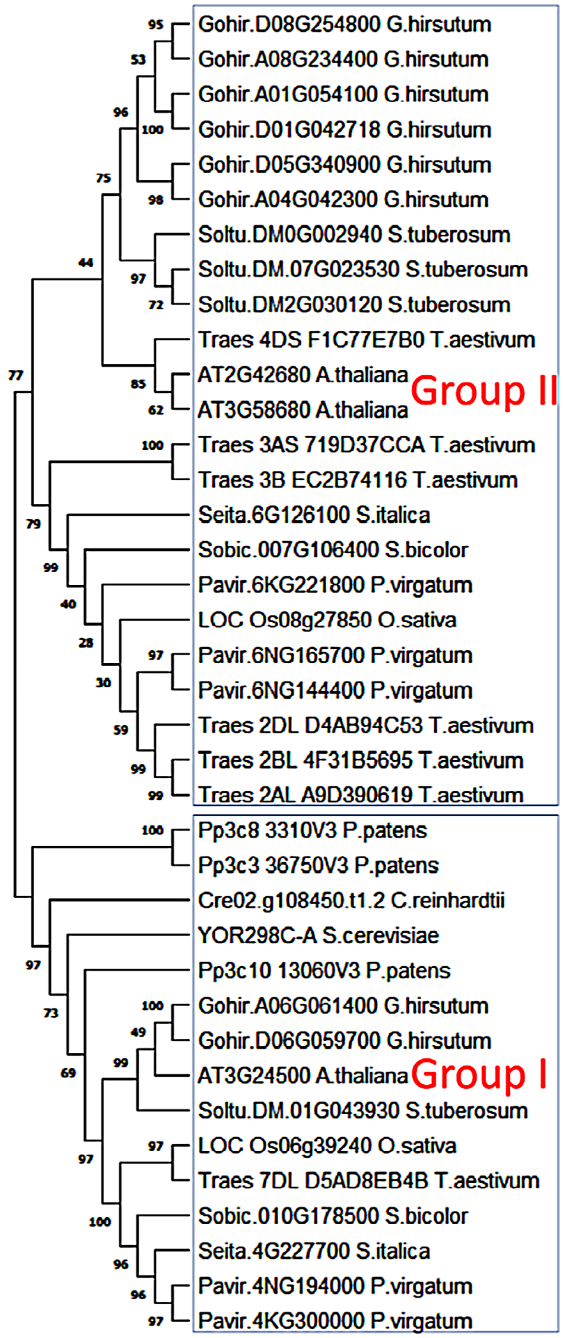

Fig. 2 Phylogenetic tree of MBF1s from eleven organisms including yeast and ten plant species. Amino acid sequences of MBF1s were aligned using the MUSCLE program and the tree was built by the neighbor-joining method using MEGA11 program (1000 bootstraps). Bootstrap values (%) are shown near the branches. Phylogenetic analysis shows 2 distinct groups of MBF1s. The Phytozome v13 accession numbers of MBF1s of different plant species are mentioned in the tree. MBF1 amino acid sequences of the following species were included: *Gossypium hirsutum* (8), *Oryza sativa* (2), *Arabidopsis thaliana* (3), *Physcomitrium patens* (3), *Sorghum bicolor* (2), *Chlamydomonas reinhardtii* (1), *Panicum virgatum* (5), *Setaria italica* (2), *Solanum tuberosum* (4), *Triticum aestivum* (7), and *Saccharomyces cerevisiae* (1). The number written within parenthesis represents the number of MBF1 members of a species included in this analysis

(microarray and RNA-seq) data in Genevestigator was accessed. It was observed that *OsMBF1s* express in all tissues (Fig. S2). *OsMBF1b* exhibited higher transcript abundance in comparison to *OsMBF1c*. *OsMBF1c* showed differential expression in different tissues and organs while *OsMBF1b* was found to be strongly expressed in all parts and showed relatively less fluctuation. *OsMBF1c* showed more expression in callus, seedling coleoptile, and inflorescence. Both genes had the highest expression in the endosperm of rice grain. Both of them displayed higher transcript abundance in flag leaf as compared to leaf. Developmental stages specific expression analysis showed that *OsMBF1b* exhibits prominent expression throughout rice development while the expression of *OsMBF1c* gradually increases during the reproductive stages of grain filling and ripening (Fig. S3).

Analysis of temporal expression

A web-based tool Diurnal (http://diurnal.mocklerlab.org/) was used to study the temporal expression of *OsMBF1* genes. Diurnal contains data from time-dependent microarray experiments carried out on model plants. This tool also identifies the most highly correlated model for an expression pattern of a gene. Three data sets of rice were assessed LDHC, LLHC, and LDHH (Fig. S4). L and D represent photocycle of 12 h light and 12 h dark while H and C refer to coupled thermocycle of heat (31ºC for 12 h) and cold (20ºC for 12 h), respectively. It was observed that *OsMBF1* genes display diurnal rhythmic expression within a period of 24 h. Transcript levels of *OsMBF1c* and *OsMBF1b* showed a peak at 4 and 9 h, respectively during daytime under LDHC condition (Fig. S4, left panel). In the case of a modified cycle (LLHC) having continuous light for 24 h along with 12 h 31ºC/12 h 20ºC thermocycle (Fig. S4, middle panel), the phase (peak time) of *OsMBF1c* did not change while that of *OsMBF1b* showed a minor shift to 7 h. Interestingly, transcript levels of both *OsMBF1* genes increased significantly under LLHC condition in comparison to that of LDHC. In case of another modification of cycle (LDHH) having a continuous high temperature of 31ºC with a photocycle of 12 h light/12 h dark, the phase of *OsMBF1c* and *OsMBF1b* got rescheduled at 23 and 10 h (Fig. S4, right panel). In this condition, their expression levels were lower than those under LDHC condition.

Stress-specific expression analysis

Stress-related microarray and RNA-seq data of rice were again accessed using Genevestigator. Differential expressions of *OsMBF1s* were analyzed with a filter of P-value < 0.05 and fold change cut off ≥ $\log_2 1.5$. *OsMBF1c*

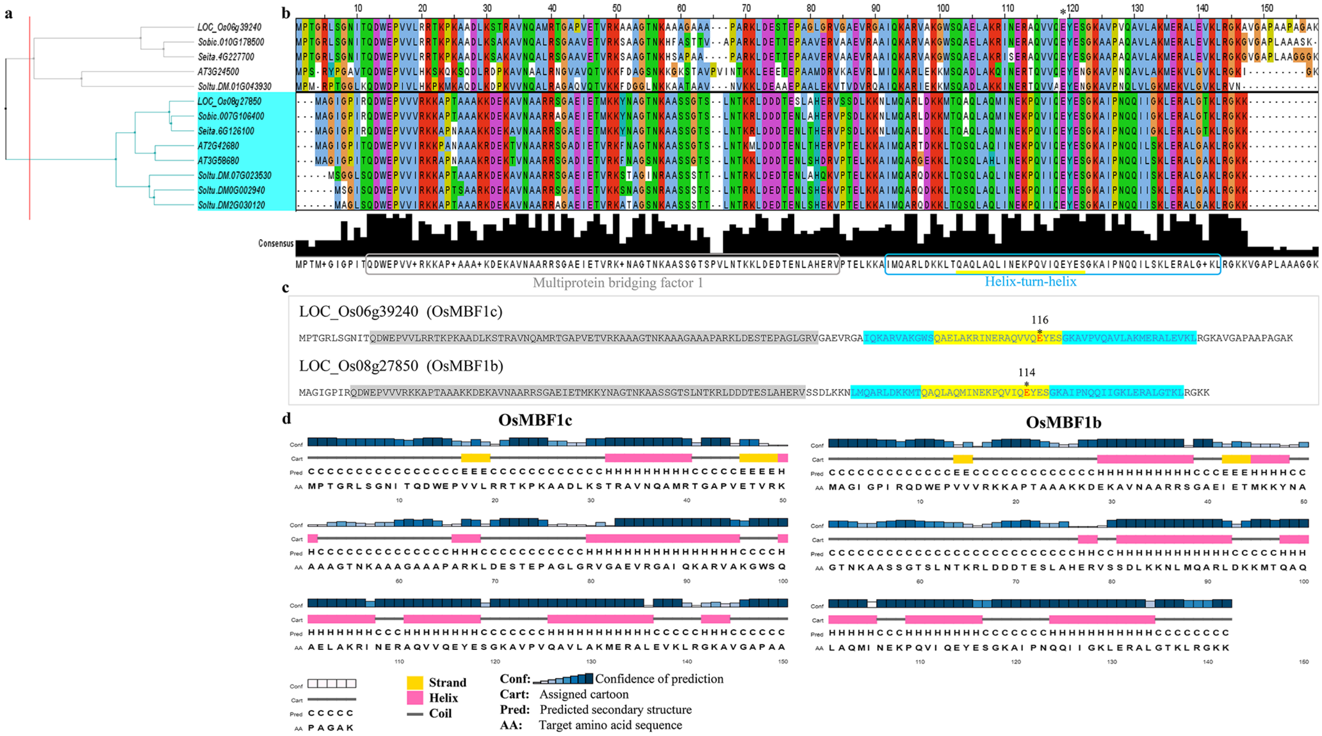

Fig. 3 Structural analysis of rice MBF1 proteins. (**a**) Phylogenic tree and (**b**) amino acid sequence alignment of MBF1s from five different plant species. The alignment was generated with the MUSCLE program and phylogenetic analysis was done by constructing an average distance tree using BLOSUM62 in the Jalview Java (Waterhouse et al., 2009). Residues are color-coded according to their conservancy. Regions of multiprotein bridging factor 1 and helix turn helix domains have been marked. (**c**) Amino acid sequences of OsMBF1s with the demarcation of the above-mentioned domains. DNA binding region within the helix turn helix domain is highlighted in yellow color and the position of a conserved glutamate residue within this region has been marked by an asterisk. (**d**) Graphical outputs of the secondary structure of OsMBF1s produced by PSIPRED-a Protein Structure Prediction Server. The predicted structure elements (E, strand; H, helix; C, randomly structured coil regions) of the target sequences are shown according to the symbols displayed in the legend. The level of confidence of prediction is shown in the figure

was found to show differential expression under several stress conditions (Table S2; Fig. S5 and S6) but *OsMBF1b* did not show such fold change in its expression level (Table S2; Fig. S7 and S8). It is noteworthy that the basal expression level of *OsMBF1b* is much higher than *OsMBF1c* probably that's why its expression did not change beyond 1.5 ($\log_2$) folds (2.83 folds in terms of absolute value). The expression of *OsMBF1c* was strongly upregulated by heat stress of different durations in different rice genotypes (Table S2; Fig. S5 and S6) at the seedling stage and got downregulated by heat recovery (Fig. S6). *OsMBF1c* also exhibited significant upregulation at the reproductive stage (Table S2). It was induced by heat in young florets of a heat-tolerant rice cultivar-996 as well as anthers of an African heat-tolerant landrace SDWG005 and an Indian aus-type landrace N22 (Table S2). Reproductive stage heat stress is more detrimental as it greatly affects rice yield. *OsMBF1c* was also induced by drought in different contrasting genotypes such as drought-tolerant Moroberekan, PSBRC28, IRAT109, and H471 as well as drought-sensitive IR64, Nipponbare, and Huang-Hua-Zhan genotypes (Fig. S6). Its expression was induced by both early (seedling stage) as well as terminal (tillering stage) droughts (Table S2). It showed strong upregulation by drought in Nagina 22 rice which is tolerant to both drought and heat (Table S2; Fig. S5). Its expression also increased in pistils and anthers under combined heat and drought in heat-tolerant Nagina 22 as well as heat-sensitive Japonica cultivar Moroberekan (Table S2; Fig. S5). Expression in anthers increased by more folds compared to pistils. Salt stress also upregulated its transcript level in both roots and shoots of seedlings of salt-tolerant Baldo and salt-sensitive Vialone Nano Japonica genotypes (Table S2; Fig. S6). It was also upregulated in shoots of salt-sensitive IR64 seedlings by salt (Table S2; Fig. S5). Its expression was found to be induced by arsenate toxicity in roots of different genotypes such as IR64, Bala, Azucena, and high-arsenic accumulating BRG12 and low-arsenic accumulating CN1646-5 genotypes (Table S2; Fig. S5). Induction was more folds in CN1646-5 in comparison to BRG12. It was also upregulated by aluminium toxicity in roots of seedlings of aluminium stress-resistant Azucena rice (Table S2; Fig. S6). Other heavy metals like Cd, Cr, and Pb also induced its expression in the roots of IR64 seedlings (Table S2; Fig. S5). *OsMBF1c* also got upregulated in alkaline stress tolerant WD20342 genotype when its seedlings were transferred into 0.5% Na_2CO_3 solution (pH 11.37) (Table S2; Fig. S6). Its expression also increased during seed germination, especially under anaerobic conditions (anoxia) (Table S2) and decreased when the aerobic condition was restored in case of Amaroo genotype which is tolerant to anaerobic conditions during the imbibition stage (Fig. S5 and S6). Under submergence stress, *OsMBF1c* showed genotype-dependent and development stage-dependent variation in its expression. It was upregulated in a submergence-intolerant Japonica cultivar M202 (Table S2; Fig. S5) which at the *Sub1* locus lacks the *Sub1A* gene (Jung et al., 2010). *OsMBF1c* was also upregulated when 6-leaf-stage C9285 plants were submerged however it was downregulated when younger C9285 plants were subjected to submergence stress (Fig. S6). C9285 is an Indica deepwater cultivar. Interestingly, once C9285 plants reach the 6-leaf-stage, their internodes can elongate in response to submergence however younger C9285 plants cannot respond to submergence by internode elongation (Minami et al., 2018; Hattori et al., 2009). *OsMBF1c* was upregulated by submergence in non-deepwater (submergence-intolerant) Japonica variety Taichung 65 which shows little internode elongation under deepwater conditions (Table S2; Fig. S6). In cold-sensitive IR29, the transcript level of *OsMBF1c* decreased due to cold stress at 4°C for 48 h but when the seedlings were transferred back to 29°C for 24 h, expression again increased (Fig. S5). Cold-tolerant Li-Jiang-Xin-Tuan-Hei-Gu (LTH) landrace showed the opposite behavior (Fig. S5). *OsMBF1c* transcript level was downregulated by phosphorus and iron deficiency (Fig. S5). Its expression increased again upon phosphorus supply (Fig. S6). It showed upregulation during various biotic stresses caused by rice blast fungus *Magnaporthe grisea*, nymphs of brown planthopper *Nilaparvata lugens Stål*, *Xanthomonas campestris* (pathogen of bacterial leaf spot), *Magnaporthe oryzae* pv. *oryzae* (causative agent of rice blast), *Xanthomonas oryzae* pv. *oryzicola* (causative agent of rice leaf blight), *Rhizoctonia solani* (a necrotrophic fungus causing rice sheath blight) and rice dwarf virus. The highest increase was found to be 91.43 folds caused by *Magnaporthe grisea* (Table S2; Fig. S5 and S6). Elicitor such as cellulase (ClsA) secreted by bacterial pathogen as well as wounding also induced its transcript level (Table S2; Fig. S5). Interestingly, *OsMBF1c* also exhibited inducibility by salicylic acid which mediates plant defence response upon pathogen infection (Table S2; Fig. S5). *Trans*-zeatin also induced it.

As mentioned earlier the expression of *OsMBF1b* was not observed to change by 2.83-fold (absolute value) or more by perturbations (Table S2; Fig. S7 and S8). Expression of *OsMBF1b* was triggered by heat stress in different rice genotypes (Table S2; Fig. S8) and came down during heat recovery phase (Fig. S8). Its expression was by upregulated in Nipponbare genotype by drought imposed under short and long-day conditions (Table S2). However, in contrast to *OsMBF1c*, its expression was decreased by drought in seedlings of IR64 and Nagina 22 rice (Fig. S7). Its expression was slightly downregulated under 48 h of 4ºC cold stress in cold-sensitive IR29 and cold-tolerant Japonica landrace Li-Jiang-Xin-Tuan-Hei-Gu and increased again when seedlings are shifted to 29ºC temperature (Fig. S7). Its transcript level was downregulated in roots under submergence stress in deepwater Indica cultivar C9285 as well as in a non-deepwater Japonica variety Taichung 65 (Fig. S8). Its expression was also decreased in roots by phosphorus deficiency (Fig. S8). *OsMBF1b* showed upregulation by infection of *Xanthomonas oryzae* pv. *oryzae* in susceptible Nipponbare genotype (Table S2; Fig. S7).

Interaction study

Analysis of protein-protein interaction of OsMBF1c and OsMBF1b was done using STRING vs11.5 (https://string-db.org; Szklarczyk et al., 2017) (Fig. S9). An interaction score cut-off of 0.7 was set to identify interactors with high confidence. The list of top ten interactors of OsMBF1c and OsMBF1b has been provided in table S3. Interestingly, six of them are common partners. Identified interactors included different proteins like HSFs, HSPs, activators of HSPs, peptidyl-prolyl cis-trans isomerases (cyclophilin-40), regulator of chaperone, and proteins having Fes1 and CS domains which are required to bind HSPs. Interaction scores were found to be 0.87–0.97 and 0.85–0.92 in the case of OsMBF1c and OsMBF1b, respectively which indicated very high confidence of interactions. As obvious from the functions of the interactors mentioned in table S3, this analysis strongly supported the role of OsMBF1s in heat responses and maintenance of proper structure and folding of proteins during high temperature. Since members of the basal transcription machinery (with which MBF1 proteins are known to interact) were not found in the list of top ten interactors, the interaction analysis was extended with a focus on the identification of proteins that are part of the physical subnetwork (physical complex) to which OsMBF1s belong. Applying an interaction score cut-off of 0.7 in STRING database, physical subnetworks of both OsMBF1 proteins were predicted (Fig. S10). It was found that both OsMBF1 proteins interact with three common TATA-binding proteins (Table S4).

Co-expression analysis

Co-expression of *OsMBF1s* with other rice genes was analysed in three conditions i.e. perturbation (different stresses), development (different developmental stages), and anatomy (different tissues) using the Genevestigator tool (Table S5). A list of the top ten co-expressed genes with each *OsMBF1* in these three conditions is provided along with Pearson correlation coefficient. Under perturbation condition, most of the genes that co-express with *OsMBF1s* are related to heat response. Interestingly, a gene (LOC_Os02g04650) encoding activator of 90 kDa heat shock protein ATPase homolog co-expressed with both *OsMBF1s* and it was also predicted to be the interacting partner of both OsMBF1s. Under different development stages, all top ten co-expressed genes showed a Pearson correlation coefficient of more than 0.95 which indicates a strong level of co-expression. A few genes related to heat response showed co-expression with *OsMBF1s* under this condition. In different anatomical parts, genes belonging to different categories showed a variable degree of co-expression with *OsMBF1s.*

In-silico promoter analysis

With the aim of understanding the putative roles of *OsMBF1* genes in rice, we further carried out in-silico analysis of their promoters using PlantCARE tool to detect regulatory *cis*-elements which are associated with different processes, particularly stress response. Promoter sequences of *OsMBF1* genes were retrieved from RGAP database. 550 bp sequence upstream to transcriptional initiation site was retrieved for both genes. 5'UTR region was also included in promoter analysis as 5'UTR is known to contain regulatory elements (Wu et al., 2014; Kim et al., 2021). Promoter analysis showed the presence of stress-related elements together with hormone-responsive elements in the putative promoter sequences of both genes. Promoters of *OsMBF1c* (Fig. S11; Table S6) and *OsMBF1b* (Fig. S12; Table S7) contained 18 and 19 types of *cis*-acting elements, respectively and among them 8 (TATA-box, CAAT-box, Unnamed__1, Unnamed__4, MYB, Myb-binding site, WRE3, STRE) were common. TATA-box and CAAT-box are basic elements which are required for transcription. The function of unnamed elements is unknown. Elements known as MYB and Myb-binding site are required for drought inducibility while WRE3 (wound responsive element) is related to response to biotic stress. It is noteworthy that both genes contained 2 copies of the stress-responsive element STRE in their promoter. One copy was found to be present the in 5'UTR region of both genes (Fig. S11 and S12). In *Arabidopsis*, STRE is reported to serve as a binding site for HsfA1a (Guo et al., 2008). Both promoters also contained different light-responsive elements (I-box in *OsMBF1c* promoter and AE-box, Box 4, and G-Box in *OsMBF1b* promoter). Elicitor-responsive elements (box S, CCGTCC motif, and CCGTCC-box) and methyl jasmonate responsiveness elements (CGTCA-motif and TGACG-motif) were found in the promoter of *OsMBF1c* gene. It also contained as-1 (activation sequence-1) element which is activated by salicylic acid through oxidative species (Garretón et al., 2002). *OsMBF1b* promoter contained *cis*-acting regulatory elements ARE and GC-motif involved in anoxic (anaerobic) induction. One abscisic acid responsiveness element (ABRE) was found in *OsMBF1b* promoter. The presence of these regulatory elements suggested that the promoters of *OsMBF1* genes might be responsive to different biotic and abiotic stresses, and hormones.

Plantpromoter db version3.0 tool (Hieno et al., 2014) was also used to scan *cis*-acting elements and to identify core promoter structure of both genes (Fig. S13 and S14). It also led to the identification of transcription start site, TATA Box, Y (pyrimidine) patch, and regulatory element groups (REG). Their sequences, genomic coordinates, and position with respect to the start codon are provided in tables S8 and S9 for *OsMBF1c* and *OsMBF1b*, respectively. REG is a group of regulatory elements. In the promoter of *OsMBF1c* (Fig. S13, table S8), all three REGs were found to possess GCCCA element (on + or - strand) known as 'Element II of *Arabidopsis* PCNA-2, cell cycle/meristematic expression'. It is necessary for meristematic expression in *Arabidopsis* (Trémousaygue et al., 2003). Interestingly, REG1 contained a motif AACAAAC which is a type of AACA motif which regulates endosperm-specific expression. It occurs in rice glutelin genes (Kawakatsu & Takaiwa, 2010). REG3 contained a motif AACCG(G/A) known as overlapping GT1 box. Its function is not known.

In the case of *OsMBF1b* (Fig. S14, table S9), which has very high level of basal expression, TATA Box and Y patch were not detected. However, three putative REGs were identified. REG 1 and 2 contained motifs of unknown function. REG 3 contained TGAC motif. A transcriptional factor OsWRKY71 binds to TGAC-containing W-box to repress gibberellin signaling in aleurone cells (Zhang et al., 2004; Xie et al., 2005). REG3 also contained GCCCA motif called 'Element II of *Arabidopsis* PCNA-2, cell cycle/meristematic expression'.

Analysis of transposon insertion in *OsMBF1* genes

The occurrence of transposon elements (TEs) in the genomic sequences of *OsMBF1s* was analysed using RTRIP (Rice Transposon Insertion Polymorphism) database (Liu et al., 2020). This database provides information about TEs in 3000 rice accessions. It was revealed that *OsMBF1c* gene

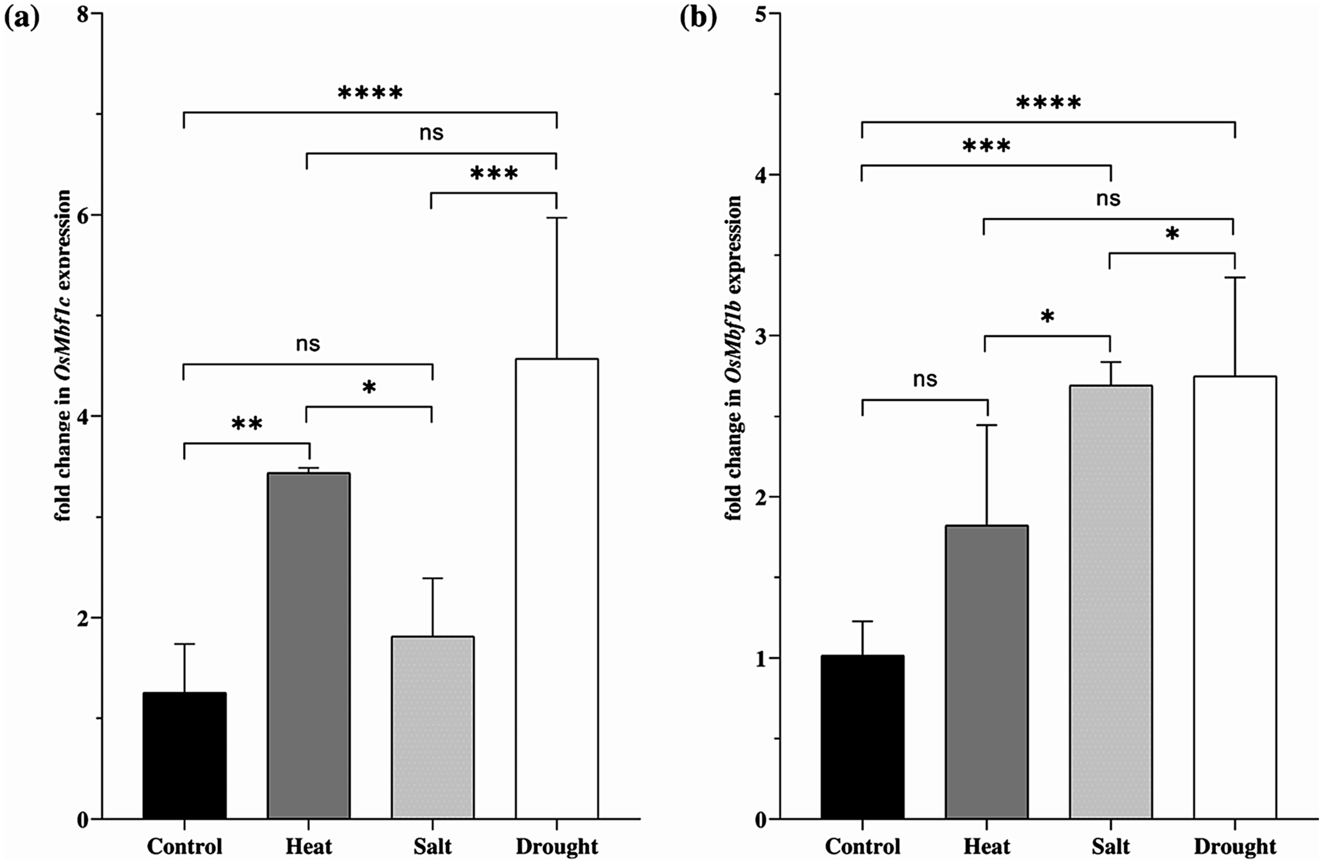

Fig. 4 Expression of *OsMBF1* genes under different stresses. Transcript abundance of (**a**) *OsMbf1c* (**b**) *OsMbf1b* genes in IR64 rice seedlings under heat (42 °C), 200mM NaCl, and 25% PEG-6000 induced drought stress. Asterisks represent significant differences in different groups based on Tukey's HSD test ($P \leq 0.05$) while 'ns' indicates non-significant change

doesn't contain any transposon. However, *OsMBF1b* gene was found to harbor transposon insertion (Stowaway50 or MITE-adh, type D) in its second intron in 42% of genotypes (Table S10). As shown in Fig. 1, *OsMBF1b* has three introns. The second one is the longest among them having 1648 bp. TE was detected to be present in either + strand (Stowaway50, length 174 bp) or – strand (MITE-adh, type D, length 154 bp) of DNA at the insertion site. Both of these TEs belong to Tc1/Mariner superfamily of transposon and share sequence similarity. Since TEs were present in the non-coding region, they may not affect the normal function of *OsMBF1b*. The absence of TE in *OsMBF1c* and its presence in the intronic region of *OsMBF1b* indicate a significant role of these genes in rice. Some of the heat-tolerant genotypes such N22 and OS4 including Nipponbare did not contain TE in *OsMBF1b* gene while other heat-tolerant ones like Carreon, Dular, Bg90-2, and Milyang23 were found to contain it.

qRT-PCR analysis of expression of *OsMBF1* genes under different stresses

To check the result of stress-related in-silico expression analysis of *OsMBF1* genes, seedlings of IR64 rice were subjected to six hours long exposure to heat, salt, and drought (Fig. 4). It caused upregulated expression of either one or both *OsMBF1* genes. *OsMBF1c* expression was significantly upregulated by drought (3.6 fold) and heat (2.7 fold) while there was non-significant change under salt stress (Fig. 4a). Significant upregulation in the expression of *OsMBF1b* was observed under drought (2.7 fold) and salt stress (2.6 fold) while heat stress-induced upregulation was found to be non-significant (Fig. 4b).

Heat stress-related expression analysis of *OsMBF1* genes was further investigated in two contrasting genotypes Nagina 22 (heat tolerant) and IR64 (heat sensitive). Their seedlings were grown under controlled conditions in a plant growth chamber. Twenty-one days old seedlings were subjected to 44°C heat stress for 12 h followed by a recovery period for 24 h at 28 ± 2 °C. Expression levels were analysed in shoots by qRT-PCR (Fig. 5). *OsMBF1c* showed

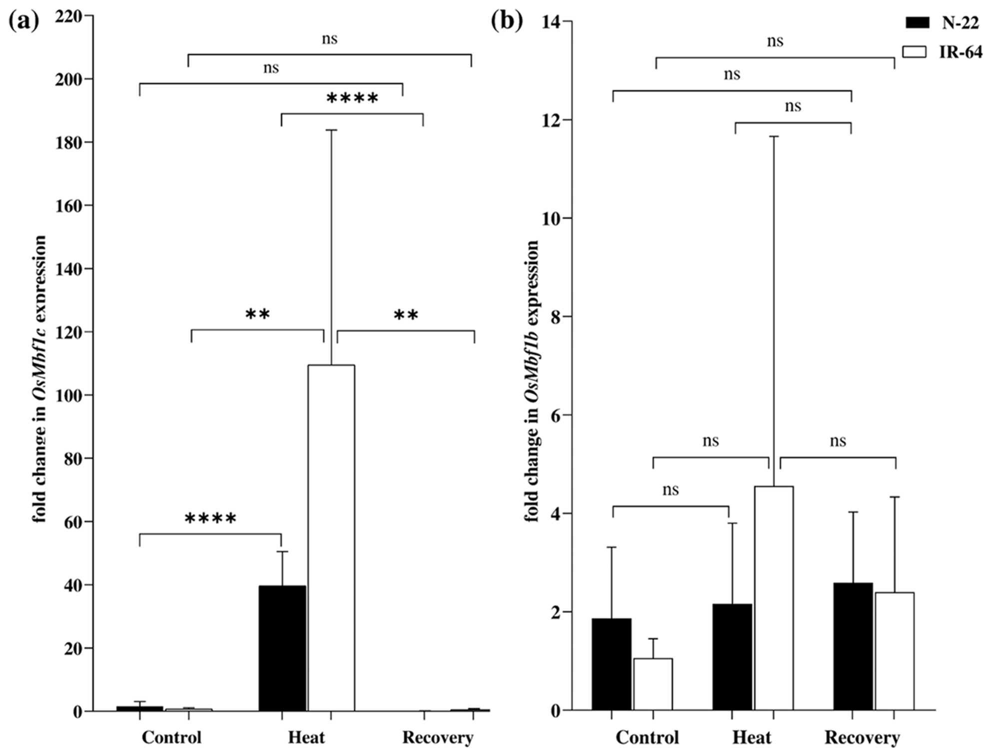

Fig. 5 Differential expression of (**a**) *OsMBF1c* and (**b**) *OsMBF1b* genes in two contrasting rice genotypes, Nagina 22 (N22) and IR64. Asterisks represent significant differences between means (based on Tukey's HSD test $P \leq 0.05$) as shown in the graphs. ns, nonsignificant

more than 35 and 100 fold upregulation in N22 and IR64 genotypes, respectively (Fig. 5a). Its expression came down to basal level (similar to that under control condition) after recovery. The transcript level of *OsMBF1b* didn't change significantly in both genotypes upon imposition of heat stress and recovery after it (Fig. 5b).

Effect of heat, salt, and drought stress on IR64 rice seedlings

Growth parameters like shoot and root length, seedling fresh and dry weight, and stress-related physiological parameters like relative water content, electrolyte leakage, and contents of photosynthetic pigments were analyzed to examine the effect of heat (42°C), salt (200 mM), and PEG-6000 (25%) induced drought stress. These abiotic stresses significantly affected all the aforementioned parameters (Figs. 6 and 7). A representative image of stressed seedlings is shown in Fig. 6a. Shoot length and root length showed an almost equal degree of decrease in heat, salt, and drought-stressed seedlings when compared to control ones (Fig. 6b and c) while fresh weight showed a greater decrease in salt and drought-stressed seedlings than the seedlings under heat stress (Fig. 6d). Dry weight was also significantly decreased under all the stresses (Fig. 6e). Electrolyte leakage was estimated to quantify the extent of cellular membrane damage in rice seedlings under these stresses (Fig. 6f). Salt stress induced the highest electrolyte leakage (48.37%) followed by heat (32.17%), and drought stress (28.51%) compared to control group (12.41%). Relative water content in control seedlings was observed to be 88.72% which was greatly affected by drought (65.75%) followed by salt (69.47%), and heat (78.32%) (Fig. 6g). Electrolyte leakage and relative water content in the salt-treated seedlings were significantly different from the heat and drought-treated seedlings (evident from multiple comparisons not shown in graphs). However, for the rest of the abovementioned parameters, the difference between heat, salt, and drought-treated seedlings was non-significant.

A significant decrease was observed in the content of photosynthetic pigments under all the studied stresses (Fig. 7). Chlorophyll a content was greatly reduced under salt stress (~971 µg/g FW) then drought (~1168 µg/g FW) and heat stress (~1300 µg/g FW) when compared to the control condition (~1565 µg/g FW) (Fig. 7a). Chlorophyll b content was the lowest under drought (~296 µg/g FW) followed by salt (~313 µg/g FW) and heat stress (~363 µg/g FW) as compared to the control condition (~423 µg/g FW) (Fig. 7b). Chlorophyll b content under heat was significantly higher than under drought stress. The contents of total chlorophyll and carotenoid followed a similar fashion under these stresses as that of chlorophyll a (Fig. 7c and d). They were lowest under salt stress (~1285 and 212 µg/g FW, respectively) followed by drought (~1465 and 232 µg/g FW) and heat stress (~1664 and 243 µg/g FW) in comparison to control (~1989 and 283 µg/g FW) (Fig. 7c and d). Total chlorophyll contents between heat and salt, and salt and drought groups were also statistically significant.

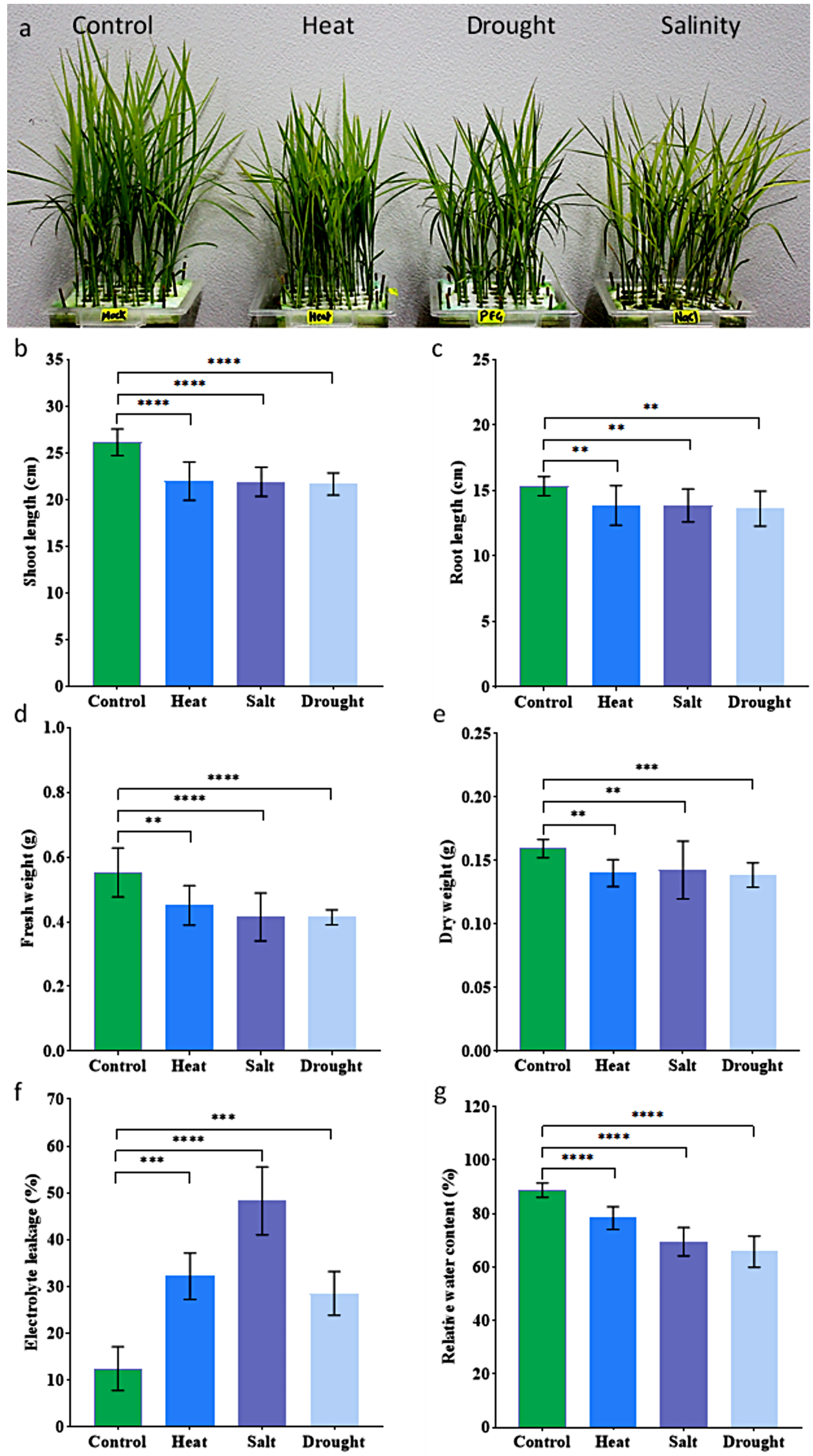

Fig. 6 Effect of heat (42°C), salt (200mM NaCl), and drought (25% PEG-6000) on growth and physiological parameters of rice seedlings. (**a**) IR64 seedlings subjected to different stresses. (**b**) Shoot length (SL), (**c**) root length (RL), (**d**) seedling fresh weight (FW), (**e**) seedling dry weight (DW), (**f**) electrolyte leakage (EL), and (**g**) relative water content (RWC). Asterisks indicate significant differences (*, $p \leq 0.05$; **, $p \leq 0.01$; ***, $p \leq 0.001$; ****, $p \leq 0.0001$) according to Tukey's test (for SL, RL, DW, and EL) and Games-Howell test (FW and RWC)

Discussion

Rice was found to contain two *MBF1* genes (Fig. 1). Till date, genome-wide identification of MBF1 members has been carried out in only two plant species i.e. *Arabidopsis* and tomato (Tsuda et al., 2004; Zhang et al., 2020). Tomato contains 5 *MBF1* genes while *Arabidopsis* has 3 genes as mentioned before. In *Caenorhabditis elegans*, zebrafish, fruit fly, human, mouse, and yeast, only one *MBF1* gene is found. In this study, when sequences of MBF1 members

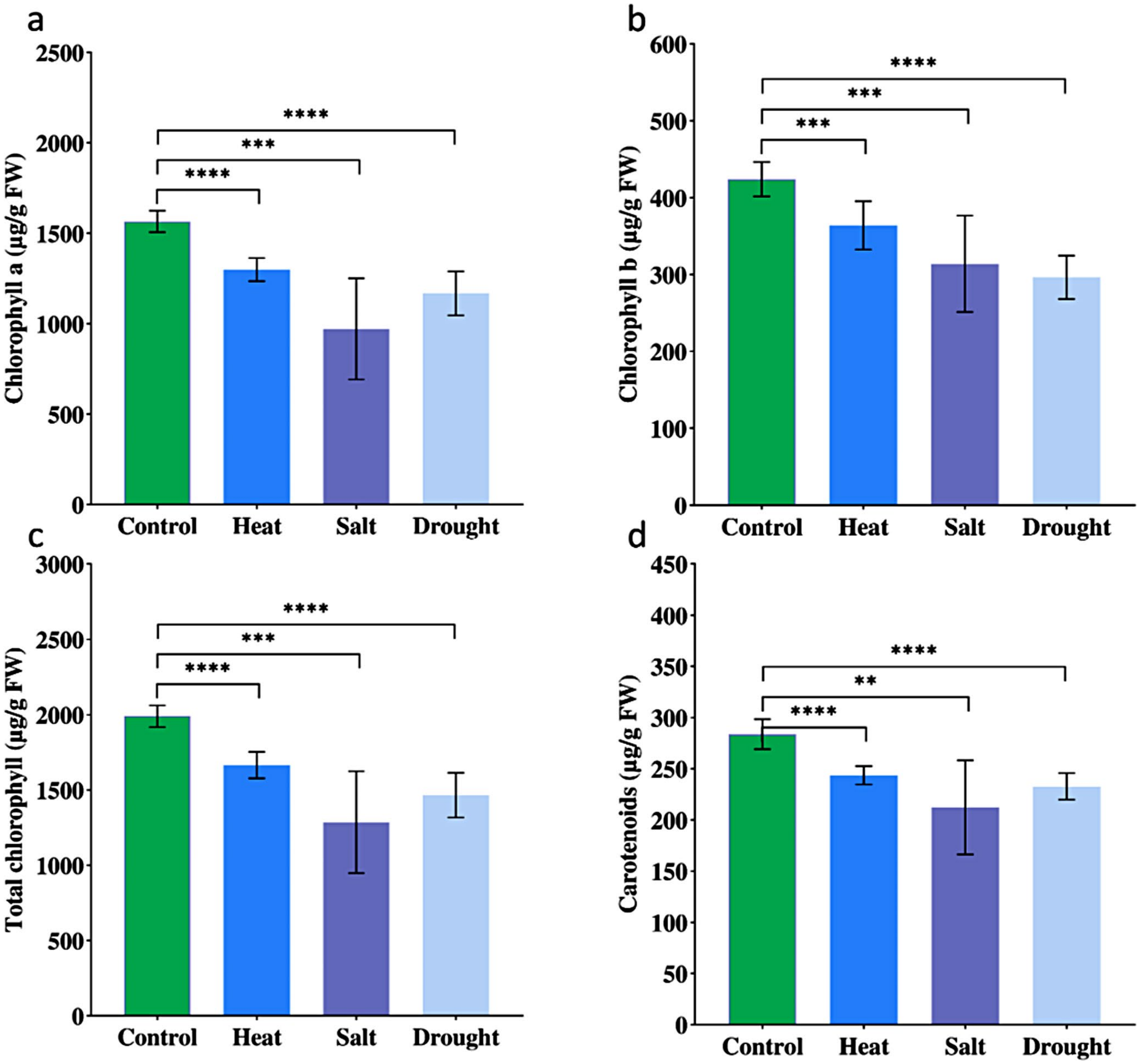

Fig. 7 Effect of heat (42°C), salt (200mM NaCl), and drought (25% PEG-6000) on photosynthetic pigments of rice seedlings. (**a**) Chlorophyll a, (**b**) chlorophyll b, (**c**) total chlorophyll, (**d**) carotenoids content. Asterisks indicate significant differences (*, $p \leq 0.05$; **, $p \leq 0.01$; ***, $p \leq 0.001$; ****, $p \leq 0.0001$) according to Games-Howell test

were retrieved from the Phytozome v13 database; different numbers of MBF1s were observed in different plant species i.e. only one in *Chlamydomonas reinhardtii* and more in plants having polyploid genomes such as *Gossypium hirsutum* and *Triticum aestivum.* It seems that MBF1 was a single gene family which got expanded in plant genomes during the course of evolution. *OsMBF1* genes show a similar structure to their counterparts in *Arabidopsis*. Like *OsMBF1c*, its homolog *AtMBF1c* (At3g24500) also lacks intron while *OsMBF1b* and its homologs *AtMBF1a* (At2g42680) and *AtMBF1b* (At3g58680) contain three introns (Fig. 1). Phylogenetic analysis revealed grouping of MBF1s into two main clusters (Fig. 2). In consistency, Tsuda and Yamazaki (2004) also found two groups of plant MBF1s. OsMBF1c and AtMBF1c belong to group I while OsMBF1b, AtMBF1b and AtMBF1a are members of group II. Both OsMBF1s contained conserved MBF1 and HTH_3 domains at N and C terminals, respectively. Differences were observed between them (Fig. 3b). The secondary structure of these two domains determines the function of MBF1 protein. In both OsMBF1s, the structure of the C-terminal domain was well-folded while the N-terminal domain did not form a well-ordered structure (Fig. S1). The C-terminal domain is more conserved across species (Ozaki et al., 1999). OsMBF1s were found to contain α-helixes (Fig. 3d and S1) which are also found in MBF1s from other species (Zhao et al., 2022). MBF1 domain is reported to interact with transcription factors (Tsuda et al., 2004) while HTH _3 domain binds TBP (Ozaki et al., 1999; Liu et al., 2007). In yeast, aspartic acid residue at 112 position is important for interaction with TBP (Takemaru et al., 1998). However, OsMBF1s and all MBF1s from other plant species included in this study were found to contain glutamate in place of aspartate (Fig. 3b and c). It has been reported that plant MBF1 proteins contain this amino acid change but it does not hamper their interaction with TBP (Tsuda et al., 2004; Liu et al., 2007). Physical subnetwork analysis, done in this study, also indicated the interaction of both OsMBF1 with three TBPs (Fig. S10; Table S4). OsMBF1 proteins were predicted to be nuclear-localized. However, OsMBF1c was predicted to be a cytoplasmic protein by WOLFPSORT. It indicates that OsMBF1c might shuttle between cytoplasm and nucleus. Its homolog in *Arabidopsis*, AtMBF1c is present in both nucleus and cytoplasm (Sugikawa et al., 2005). Nevertheless, when heat stress is imposed, it moves from the cytoplasm into the nucleus (Suzuki et al., 2008; Kim et al., 2015). A similar observation has been reported in the case of a wheat MBF1 protein (Qin et al., 2015).

Like *OsMBF1s*, *AtMBF1s* express in all tissues (Tsuda et al., 2004). The omnipresent expression of *OsMBF1s* (Fig. S2 and S3) suggests that they are essential for the normal growth and development of rice plants. In *Arabidopsis*, *MBF1* genes regulate leaf cell expansion, ploidy, and plant size (Hommel et al., 2008; Tojo et al., 2009). The absence of transposons in the *OsMBF1c* gene or their presence in the intronic region of *OsMBF1b* (Table S10) also indicates that *OsMBF1s* are essential genes and mutation in them might affect rice growth and survival. Nevertheless, the prominent expression of *OsMBF1s* in the endosperm (Fig. S2) suggests that they play an important role in seeds. The presence of an endosperm-specific element (AACA) in the putative promoter of *OsMBF1c* (as detected by the plantpromoter db tool) might be responsible for its highest transcript level in the endosperm. *Arabidopsis* homologs also exhibit similar tissue and development-specific expression pattern as indicated by their analysis using Genevestigator (Data not shown). In *Arabidopsis*, mutations in *AtMBF1* genes enhance the dormancy of seeds (Di Mauro et al., 2012). *AtMBF1c* is suggested to be associated with the regulation of ABA degradation during seed germination. However, no report elucidating MBF1 function in rice seeds is available yet. Recently, Sano et al. (2022) found that OsMBF1c protein significantly accumulates in rice embryos during germination. Interestingly, TGAC motif, the binding site of OsWRKY71, was detected by plantpromoter db tool in the putative promoter of *OsMBF1b*. However, it was not found in *OsMBF1c* promoter. OsWRKY71 blocks gibberellic acid signaling (Zhang et al., 2004). Expression of *OsWRKY71* is decreased by gibberellic acid (seed germination promoter) (Zhang et al., 2004) and induced by abscisic acid (seed dormancy promoter) in aleurone cells of rice seeds (Xie et al., 2005). It seems that during seed germination, gibberellic acid releases the suppression of the *OsMBF1* expression caused by OsWRKY71. Subsequently, OsMBF1 protein accumulates and causes regulated degradation of ABA. Probably, thus a balance of required phytohormones leads to the activation of genes essential for seed germination. Further research is required to find out the role of OsMBF1s in rice seeds. It is also noteworthy that *OsMBF1c* expression is highly induced in coleoptiles during seed germination under anaerobic conditions and its expression decreases when seeds are shifted to aerobic conditions (Fig. S5 and S6).

The analysis of temporal expression of *OsMBF1s* suggests that thermocycle strongly affects their rhythmic pattern (Fig. S4). It seems that it is the heat during the daytime which induces their transcript levels as evident by the comparison of their expression patterns under LDHC and LLHC conditions (Fig. S4). It was also observed that continuous light (LL) increases their expression as compared to the alternate light/dark cycle (LD). The presence of several light-responsive *cis*-acting elements might be responsible for this observation (Fig. S11 and S12; Tables S6 and S7). Stress-related genes are known to exhibit diurnal rhythm (Soni et al., 2013; Singh et al., 2015). Regarding the stress-specific expression analysis, a summary of the behaviour of *OsMBF1* genes is mentioned in Table 2. Since *OsMBF1b* expresses strongly in all tissues even under non-stress condition, it does not show much fluctuation in its transcript level under stressful conditions (Table S2 and 3; Fig. S7). qRT-PCR analysis also confirmed this fact as *OsMBF1b* did not show a significant change in its transcript level under heat stress as well as recovery (Figs. 4 and 5b) Stress-tolerant genotypes are known to possess higher levels of transcript abundance of stress-related genes even in the absence of stress (Gong et al., 2005; Kumari et al., 2009). So non observation of more than 1.5 fold ($\log_2$ FC) change in *OsMBF1b* expression level under perturbation conditions (Fig. S7 and S8) during in-silico analysis or by qRT PCR, does not rule out its role in stress response. In a study related to genome-wide expression profiling covering the entire life cycle of rice, *OsMBF1b* was also found to be among the top ten genes which exhibit constitutive high and stable expression in all the tissues (Wang et al., 2010). *OsMBF1b* expression was detected (data not shown) to be as high as those of known housekeeping genes in rice such as *ACT11*, *eEF-1a*, *UBC,* and *GAPDH* as reported by Jain et al. (2006). It is not

Table 2 Differential upregulation (with a cut-off value $\geq \log_2 1.5$ in terms of fold change, *P*-value > 0.05) of *OsMBF1* genes under various stresses

Gene	Heat	Drought	Drought + Heat	Submergence	Salt	Cold	Heavy metals	Hormone	Biotic stress
OsMBF1c	Yes	Yes	Yes	Differential regulation (Genotype dependent)	Yes	Differential regulation	Yes	Yes (by salicylic acid and *trans*-zeatin)	*Magnaporthe grisea*, *Xanthomonas oryzae pv. oryzae*, *Xanthomonas oryzae pv. oryzicola*, *Xanthomonas campestris pv. vesicatoria*, *Rhizoctonia solani*, rice dwarf virus
OsMBF1b	Yes	Differential regulation (Genotype dependent)	No	Down-regulation	No	Down-regulation	No	No	*Xanthomonas oryzae pv. oryzae*

surprising that Wang et al. (2016) found *OsMBF1b* suitable to be used as a reference gene (in combination with other ones) for normalization of expression data of qRT-PCR studies related to hormonal (6BA, IAA, GA, SA, and ABA) treatment to rice plants.

In diverse plants such as *Pyropia yezoensis*, *Polytrichastrum alpinum*, *Retama raetam*, *Arabidopsis,* and wheat; *MBF1s* (belonging to group I) have been observed to be strongly upregulated by heat stress (Pnueli et al., 2002; Tsuda & Yamazaki, 2004; Suzuki et al., 2005, 2008; Uji et al., 2013; Qin et al., 2015; Alavilli et al., 2017). We also observed heat-induced upregulation of the *OsMBF1c* gene (Fig. 5A). The presence of two STRE (AGGGG) elements in the putative promoter of *OsMBF1c* gene (as detected by in-silico promoter analysis; Fig. S11; Table S6) might be responsible for its heat inducibility. Although this element was found to be present in *OsMBF1b* promoter (Fig. S12 and Table S7); its expression was not changed significantly under heat stress (Figs. 4 and 5B) AGGGG motif is a heat-responsive element and it has been experimentally validated that HSF binds it (Guo et al., 2008). Similar to *OsMBF1c* (Fig. 5A, Fig. S5 and S6), the expression of *AtMBF1c* in *Arabidopsis* gets decreased under heat recovery (Tsuda & Yamazaki, 2004). Overexpression of *MBF1* is reported to enhance heat tolerance in yeast, rice, *Arabidopsis,* and ryegrass (Suzuki et al., 2005; Qin et al., 2015; Huang et al., 2022). It is important that MBF1 also helps in overcoming the detrimental effects of reproductive stage heat stress on yield. Overexpression of *TaMBF1c* is reported to increase the spikelet fertility of rice from 3 to 12% under heat stress conditions compared to the wild type (Qin et al., 2015). Not only rice transcriptome data analysis but also protein-protein interaction prediction carried out in this study (Fig. S9; Table S3) indicates an important role of *OsMBF1s* in heat tolerance in rice. HSFs transcription factors are known to induce the expression of *HSPs* genes while AHSAs (activator of heat shock protein ATPase) stimulate ATPase activity of HSPs proteins (Lotz et al., 2003). In the current study, OsMBF1s were predicated to interact with both HSFs and AHSAs in addition to other proteins involved in heat response (Fig. S9; Table S3). It indicates a versatile role of OsMBF1s with respect to the regulation of heat response at the transcriptional level as well as at the post-translational level. Interestingly, a recent finding in wheat supports this observation (Tian et al., 2022). *TaMBF1c* imparts thermotolerance in wheat via translational regulation of mRNAs of heat shock proteins. Moreover, transgenic rice plants overexpressing *TaMBF1c* contain higher levels of transcripts of *OsHSPs* (Qin et al., 2015). In a xerophytic desert plant, *Cleistogenes songorica* heat-specific co-expression network was found to include *MBF1* along with *HSPs* (Yan et al., 2020). In this study, co-expression analysis (especially under perturbation; Table S5) also indicated *OsMBF1s* function in heat tolerance as several heat-responsible genes were found to co-express with *OsMBF1s*. *MBF1s* are reported to be involved in responses to other stresses too. *MBF1s* are induced by drought in different plants (Rizhsky et al., 2004; Tsuda & Yamazaki, 2004; Suzuki et al., 2005; Ray et al., 2011; Baloglu et al., 2014; Yan et al., 2014; Qin et al., 2015; Suresh et al., 2022). It is consistent with our qRT PCR analysis which confirmed drought inducibility of both *OsMBF1* genes in the IR64 genotype (Fig. 4). IR64 is a high-yielding mega variety of high-quality (Mackill & Khush, 2018). However, it is sensitive to salt, drought, and heat stress as indicated by the analysis of different parameters (Figs. 6 and 7). Similar findings regarding the nature of IR64 have been observed in earlier studies also (Mishra et al., 2021; Yadav et al., 2022; Anwar et al., 2024). In *Vitis vinifera*, expression of *VvMBF1* (a group I *MBF1*) increases under drought and declines upon rewatering (Yan et al., 2014). Similarly, *OsMBF1c* exhibited upregulation by drought and downregulation by rehydration (Fig. S5 and S6). Ray et al. (2011) have also reported induction of *OsMBF1c* by water-deficit stress. However, *OsMBF1b* showed genotype-dependent differential regulation under drought stress as revealed by analysis of transcriptomic data (Fig. S7 and S8). Although the ABRE element was detected in its putative promoter in the reference genome of Nipponbare (Fig. S12; Table S7), *OsMBF1b* was not observed to be induced by ABA. In a desert legume *Retama raetam*, *MBF1* was found to respond to hot and dry climate of arid region (Pnueli et al., 2002). In *Arabidopsis* and tobacco, *MBF1s* show higher transcript levels under combined heat and drought stress as compared to a single one. (Suzuki et al., 2005). Similarly in rice, *OsMBF1c* was observed to exhibit the same response (Table S2; Fig. S5).

Overexpression of *MBF1* confers salinity tolerance in plants (Zhao et al., 2019; Zhang et al., 2020). *Arabidopsis* triple mutant *mbf1abc* is overly sensitive to salt stress (Arce et al., 2010). In *Arabidopsis*, *MBF1s* have also been shown to be upregulated by NaCl treatment (Tsuda & Yamazaki, 2004; Kim et al., 2007). In this study, *OsMBF1c* showed inducibility by salt stress as revealed by analysis of publicly available transcriptomic data (Table S2 and 3; Fig. S6). However, in qRT-PCR analysis, salt inducible significant upregulation was observed for the *OsMBF1b* gene (Fig. 4) The expression of *OsMBF1b* is downregulated by cold (Fig. S7). However, *OsMBF1c* showed genotype-dependent and duration-dependent differential expression under cold stress (Fig. S5). Overexpression of group I *MBF1* genes either does not affect or decrease cold tolerance in *Arabidopsis* (Suzuki et al., 2005; Guo et al., 2014).

MBF1s are also involved in protection against biotic stresses in plants via jasmonic acid and ethylene-dependent pathways (Kim et al., 2007). Overexpression of *MBF1s* in *Arabidopsis* enhances its resistance against *Pseudomonas syringae* and *Botrytis* (Suzuki et al., 2005; Kim et al., 2007). *StMBF1* shows increased transcript abundance in potato tubers upon *Fusarium* infection (Godoy et al., 2001). It is also induced by ethylene, salicylic acid as well as wounding. *StMBF1c* is involved in the defense against bacterial wilt caused by *Ralstonia solanacearum* in potato (Yu et al., 2021). In the current study, both *OsMBF1s* were observed to be induced by biotic stresses caused by different pathogens of rice (Table S2 and 3; Fig. S5-S8). Putative promoters of both genes were also found to contain wound responsive element (Fig. S12 and S13). *OsMBF1c* was observed to be induced by wounding (Table S2; Fig. S5). *OsMBF1c* promoter possessed TGACG-motif and CGTCA-motif involved in the methyl jasmonate responsiveness. Interestingly, *OsMBF1c* was found to be induced by salicylic acid. Its putative promoter also contained as-1 element which is activated by salicylic acid via reactive oxygen species (Singh et al., 2019).

OsMBF1c was found to exhibit increased transcript levels in roots under heavy metals stress (Fig. S5 and S6). It showed genotype-dependent and development phase-dependent differential regulation under submergence stress in shoot (Fig. S5 and S6). As mentioned before, its expression was also induced during seed germination in anaerobic conditions and returned to normal upon restoration of aerobic conditions (Fig. S5 and S6). These three observations are new findings regarding the role of *MBF1s*. Till date, there is not a single report with reference to the characterization of *MBF1* in submergence, heavy metals stress and seed germination. In a tree, *Broussonetia papyrifera*, a *MBF1* gene has been reported to be significantly upregulated by Cd toxicity (Xu et al., 2019). Mercury treatment to pepper seedlings suppresses *CaMBF1* expression (Guo et al., 2014). Altogether bioinformatics data mining performed in this study and reported evidence from experiments carried out on OsMBF1 orthologs in other organisms support that *OsMBF1s* may play a prominent role in various stresses in rice, especially in heat tolerance as also confirmed by qRT-PCR analysis. This bioinformatics study also indicated new functions of rice *MBF1* genes which need to be functionally validated in the future.

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Author contributions PS conceived the idea, carried out all bioinformatic/in-silico data analysis, and prepared the manuscript. AB carried out all wet lab experiments and contributed to writing and proofreading. HR did editing and proofreading of the manuscript.

Declarations

Declaration of generative AI and AI-assisted technologies in the writing process No AI-based tool was used for writing the manuscript.

Competing interests The authors declare no conflict of interest.

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